Evaluating the performance of transformer based crowd counting model: **CCTrans**

1 Model Overview

1.1 Introduction

Challenges with previous crowd-counting models have been that crowds closer to camera have lower densities as well as larger scales and vice versa. The dramatic scale and density variations in the images are usually handled by capturing global context details. These have previously been captured by CNN-based models. But they are not perfect due to limited receptive fields. Vision transformers have been a major point of discussion for this task. The paper by *Chu et al.* focuses on simplifying the pipeline for transformer-based crowd-counting models.

1.2 Pipeline

1.2.1 2D image to 1D sequence

The model first converts the input image $I \in \mathbb{R}^{H \times W \times 3}$ into a sequence $e \in \mathbb{R}^{N \times D}$. I is divided into $\frac{H}{K} \times \frac{W}{K}$ patches each of size $K \times K \times 3$ where $H, K, 3$ are, respectively, the height, width, and channels of I. This 2D patch array is flattened into a 1D patch sequence $x \in \mathbb{R}^{N \times D}$ where $N = \frac{HW}{K^2}$ and $D = K \times K \times 3$. The sequence x is projected into an embedded sequence e using the learnable projection $f : x_i \rightarrow e_i \in \mathbb{R}^D$ $(i = 1, \ldots, N)$. In this way, spatial and channel features of the i-th image patch x_i are transformed into embedded features of the i-th embedding vector e_i.

1.2.2 Transformer backbone

The backbone of this model is a pyramid vision transformer. It captures both local and global features of this image. This is achieved mainly through two attention modules locally-grouped self-attention(LSA) and global sub-sampled attention(GSA). LSA captures local details by dividing e into sub-windows of size $k_1 \times k_2$. Multi-head self-attention is performed on these sub-windows. GSA captures global details by finding a representative element for each of these sub-windows. Multi-head self-attention is performed on this representative sequence and then up-scaled. The necessary multilayer perceptron(MLP) modules and

layer normalization modules are interleaved. The $l - th$ layer is made up as in Figure 1. Between each layer the backbone adopts sub-sampling for keys to reduce compute cost.

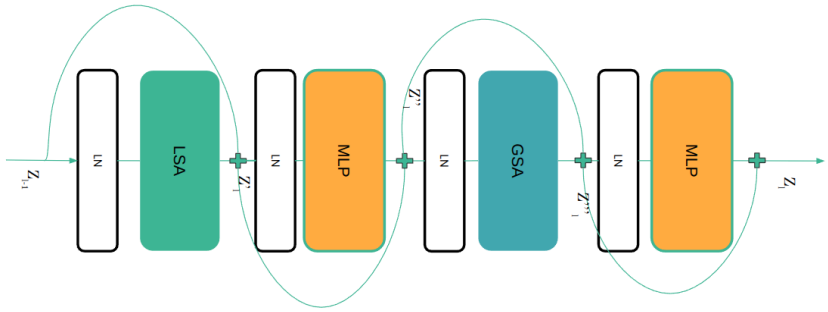

Figure 1: Transformer block with MLP, GSA, and LSA components

1.2.3 Pyramid Feature Aggregation (PFA)

Each layer generates a feature map with different features. The high-level feature map of deeper layers are too fuzzy to distinguish boundaries of different objects. To bring back these details, we up-sample the feature amps of deeper layers to $\frac{1}{8}$ size of the input image, we add these up-sampled feature maps to output a final feature map that will be used to predict the density map.

1.2.4 Regression Head

The feature map is processed with a simple regression head having 3 channels, C_1, C_2, C_3. A channel is constituted as in the Figure 2.

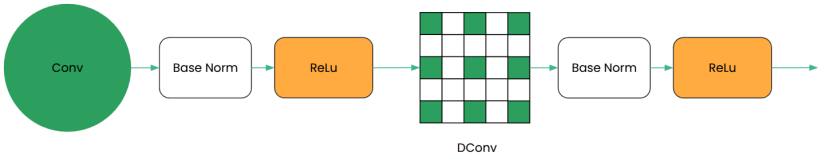

Figure 2: Regression head Channel

The main difference between the channels is the parameters for the convolution and dilated convolution modules.

1.2.5 Loss function

The loss function used by this model is given by:
$\mathcal{L}_d = L_1(P, G) + \lambda_1 \mathcal{L}_{OT} + \lambda_2 L_2(D, D')$.

where D is the predicted density map, D' is the ground truth density map, P is predicted count and G is the ground truth count. L_1 is Mean Absolute error ($|P - G|$), L_2 is Mean squared error ($\sum(D_{ij} - D'_{ij})^2$) and $\mathcal{L}_{OT}$ is the optimal transport loss (used when spatial layout matters). λ_1,λ_2 are parameters where $\lambda_1 = 0.01$ and λ_2 can be finetuned ($= 1.0$ here).

2 Results

2.1 Performance

The CCTrans model performs exceedingly well in the ShanghaiTech dataset (both parts A and B).

Dataset	**MAE**	**MSE**
ST_Part A	52.3	84.9
ST_Part B	6.2	9.9

Table 1: MAE and MSE values for ShanghaiTech dataset

The images of the ShanghaiTech Part A are randomly collected from the internet. The number of people in these images vary significantly. Part B images have dramatic intra-scene scale and density variances.

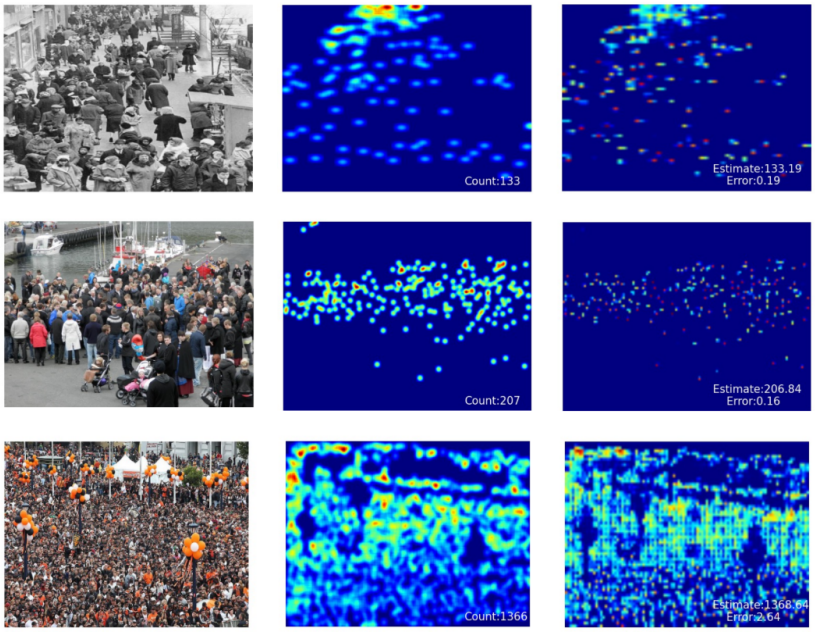

Figure 3: ShanghaiTech Part A visualisations

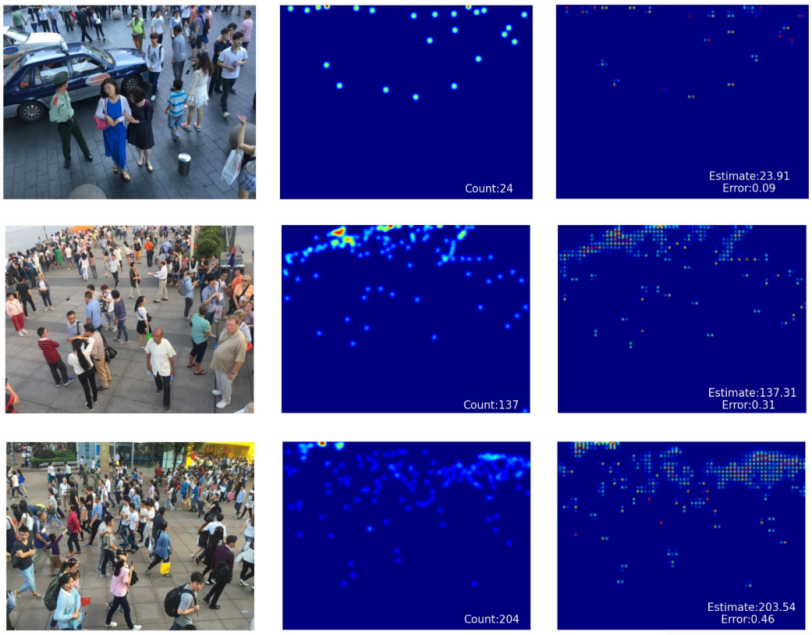

Figure 4: ShanghaiTech Part B visualisations

Visualizations for ShanghaiTech Part A images and Part B images are given in Figure 3 and Figure 4 respectively.

2.2 Computation Power

This model was trained on a single 32G Tesla V100 GPU. The training time $\approx$ 16–20 hours. The inference time for a 384×384 image $\approx 10 - 15$ fps.

3 Conclusion

In this report, we evaluated the CCTrans model, a transformer-based architecture tailored for crowd counting. By leveraging a pyramid vision transformer backbone, combined with locally-grouped self-attention (LSA) and global sub-sampled attention (GSA), CCTrans effectively captures both fine-grained local features and broader contextual information. The Pyramid Feature Aggregation (PFA) and multi-channel regression head further enhance the model's capability to generate accurate density maps. Results on the ShanghaiTech dataset demonstrate that CCTrans achieves strong performance, especially in terms of Mean Absolute Error (MAE) and Mean Squared Error (MSE), while maintaining reasonable inference speeds. This shows the promise of transformer-based methods in crowd counting, offering both accuracy and scalability with a simplified pipeline.